

Day 04 – Compute (EC2 / Virtual Machines)

Overview

Compute services allow users to run applications on virtual machines in the cloud.

Amazon EC2

EC2 provides resizable virtual servers (instances) for running applications.

Key Features

- Scalable compute
- Pay-as-you-go
- Multiple instance types
- Secure access with key pairs

Core Concepts

- Instance: Virtual server
- AMI: OS template
- Instance Type: CPU & RAM
- Security Group: Firewall

Security

- Key pair authentication
- Security group rules
- IAM roles

Azure Virtual Machines

Azure VM is similar to EC2 with flexible scaling and integration.

Examples (Real-World)

1. Host a website on EC2
2. Run a backend API server
3. Deploy ML model
4. Create testing environment
5. Run database server
6. Game server hosting
7. Batch data processing
8. DevOps CI/CD runner
9. Container hosting (Docker)
10. Remote desktop server

Learning Summary

Understood EC2, VM setup, security, and real-world usage

Workflow Diagram

